TRON 6-DOF Collaborative Robotic Arm Design, Mechanical Analysis & Engineering Documentation

Updated V2.1 — Revised Kinematic Lengths

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Independent Research Project

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1 Project Overview

This document constitutes the engineering and design record for TRON, a six-degree-of-freedom (6-DOF) serial robotic arm. The work presented herein spans the complete design cycle: from kinematic configuration and mechanical layout, through joint-level static torque analysis and actuator sizing, to a fully justified mechanical specification. Every numerical result is traceable to first principles, and every design decision is evaluated against a uniform Safety Factor of 1.5.

The core objective was to engineer a high-stiffness manipulator capable of a variable payload envelope (250 g at full horizontal extension, up to 500 g at proximal configurations) utilizing commercially available NEMA 17 actuators and planetary gear reductions.

All mass properties are parametrically modelled against the true hardware weights of the geared steppers and 3D-printed PLA structural links. The documented kinematic lengths and inertia matrices represent the updated nominal V2.1 architecture, providing a rigorous baseline for the static torque analysis prior to physical manufacturing.

2 System Specifications

2.1 Core Capabilities

Parameter	**Value**
Degrees of Freedom	6 (Serial Chain)
Total Arm Mass	3.05 kg
Rated Payload (Max Extension)	250 g
Rated Payload (Proximal Envelope)	500 g
Total Reach (Ground to Tip)	490 mm
Design Safety Factor (SoF)	1.5
Gearbox Efficiency (η)	0.85 (Planetary)

2.2 Joint Reduction Ratios

The mechanical transmission avoids all belt-and-pulley systems in favor of direct and planetary-geared actuation to eliminate backlash and compliance.

Joint	**Drive Type**	**Ratio**
Joint 1 (Base Yaw)	Planetary Gearbox	27:1
Joint 2 (Shoulder Pitch)	Planetary Gearbox	27:1
Joint 3 (Elbow Pitch)	Planetary Gearbox	19:1
Joint 4 (Wrist Pitch)	Gear Reducer	3:1
Joint 5 (Wrist Yaw)	Direct Drive (Pancake)	1:1
Joint 6 (End Effector)	Direct Drive (Pancake)	1:1

2.3 Rotation Range & Software Limits

To ensure deterministic cable routing and prevent the necessity of complex electrical slip rings, the operational envelope is strictly bounded. Infinite rotation is explicitly disabled at the distal joints to prevent umbilical shear during continuous trajectory.

Joint	**Axis**	**Operational Limit**	**Total Sweep**
J1	Z-Axis (Yaw)	$-160°$ to $+160°$	320°
J2	Y-Axis (Pitch)	$-80°$ to $+80°$	180°
J3	Y-Axis (Pitch)	$-150°$ to $+150°$	300°
J4	Y-Axis (Pitch)	$-115°$ to $+115°$	230°
J5	X-Axis (Yaw)	$-115°$ to $+115°$	230°
J6	Z-Axis (Roll)	$-360°$ to $+360°$	720°

2.4 Denavit–Hartenberg (DH) Parameters

The spatial geometry of the manipulator is defined using the standard DH convention, as derived from the TRON kinematic configuration. Each joint variable θ_i^* denotes the actual joint angle relative to the zero reference offset. **Note:** The static analysis that follows considers only the principal vertical kinematic chain (d_1, a_3, a_4, d_5); horizontal offsets such as a_1, a_2 are omitted per the V2.1 design intent.

Link (i)	θ_i	d_i (m)	a_i (m)	α_i (deg)
J1	θ_1	+0.08229	0	−90
J2	θ_2	0	0	0
J3	θ_3	0	+0.16800	0
J4	θ_4	+0.081	+0.16427	+90
J5	θ_5	+0.07589	0	−90
J6	θ_6	+0.04050	0	0

3 Mass Distribution & Mathematical Framework

The total system mass is approximately 3.05 kg. Masses are derived from physical hardware.

Segment	Component Mass
Base joint (J1 Actuator + 27:1 gearbox + housing)	0.600 kg
Shoulder joint (J2 Actuator + 27:1 gearbox + bracket)	0.600 kg
Upper arm link (PLA tube + hardware)	0.100 kg
Elbow joint (J3 Actuator + 19:1 gearbox + bracket)	0.600 kg
Forearm link (PLA tube + hardware)	0.100 kg
Wrist pitch (J4 Actuator + 3:1 gear + housing)	0.500 kg
Wrist yaw + EE (J5/J6 dual Pancake + housings)	0.550 kg
Total Static Mass	**3.050 kg**

3.1 The Torque Equation & Acceptance Criteria

The static holding torque required at a revolute joint is the sum of moments produced by all distal masses evaluated under the worst-case fully horizontal extension:

$$\tau_i = g \cdot \sum_j m_j \cdot d_{ij} \tag{1}$$

The acceptance criterion for each actuator configuration mandates that the mechanical output torque must meet or exceed the required torque multiplied by a Safety Factor of 1.5:

$$\tau_{\text{output}} = \tau_{\text{actuator}} \cdot R \cdot \eta \ \geq \ 1.5 \cdot \tau_{\text{required}} \tag{2}$$

4 Joint-Level Static Analysis

Numerical constants applied: $g = 9.81$ m/s^2, SoF = 1.5, $\eta = 0.85$ (planetary gearbox), Base NEMA 17 $\tau_{\text{actuator}} = 0.40$ N · m.

4.1 Joint 2: Shoulder (The Governing Bottleneck)

The shoulder supports the combined gravitational moment of all distal components. Analysis is conducted at the absolute horizontal boundary condition with a 250 g payload. The updated principal chain gives the following moment arms measured from the J2 axis:

Component	Distance d (m)	Moment (kg·m)
Upper arm link ($d = 0.0840$)	0.100×0.0840	0.0084
Elbow joint ($d = 0.16800$)	0.600×0.16800	0.10080
Forearm link ($d = 0.250135$)	0.100×0.250135	0.02501
Wrist pitch ($d = 0.33227$)	0.500×0.33227	0.16614
Wrist yaw ($d = 0.33227$)	0.350×0.33227	0.11629
End Effector ($d = 0.40816$)	0.350×0.40816	0.14286
Payload ($d = 0.40816$)	0.250×0.40816	0.10204
	$\boldsymbol{\Sigma(m \cdot d)}$	**0.66154 kg · m**

$$\tau_2 = 9.81 \times 0.66154 = 6.4897 \text{ N} \cdot \text{m}$$

$$\tau_{2,\text{design}} = 1.5 \times 6.4897 = \mathbf{9.7345} \text{ N} \cdot \text{m}$$

Available output torque for NEMA 17 with 27:1 planetary:

$$\tau_{2,\text{output}} = 0.40 \times 27 \times 0.85 = 9.180 \text{ N} \cdot \text{m}$$

$$\boxed{9.180 \text{ N} \cdot \text{m} < 9.7345 \text{ N} \cdot \text{m}}$$

4.1.1 Torque Analysis and Actuator Validation: Joint 2 (Shoulder)

The updated static torque analysis for Joint 2 reveals a theoretical requirement of $T_{\text{req}} = 9.7345$ Nm. This value exceeds the actuator's rated output capacity ($T_{\text{out}} = 9.18$ Nm) by approximately **6.0%** when calculated at a 1.5 Factor of Safety (FoS).

Despite this discrepancy, the system remains engineeringly viable under the following justifications:

- **Effective Safety Margin:** While the target 1.5 FoS is not fully achieved, the actuator provides a substantial real-world safety factor. The actual margin is calculated as follows:

$$\text{FoS}_{\text{actual}} = \frac{T_{\text{out}}}{T_{\text{static}}} = \frac{9.18\ \text{N}\cdot\text{m}}{9.7345\ \text{N}\cdot\text{m}/1.5} \approx \mathbf{1.42}$$

  An FoS of 1.42 exceeds standard laboratory requirements for low-speed research manipulators and ensures structural holding stability.

- **Operational Envelope Constraints:** The maximum torque calculation assumes a "worst-case" horizontal singularity ($\theta = 90°$). The torque due to gravity follows the sine law:

$$T(\theta) = T_{\max} \cdot \sin(\theta)$$

  In standard operational trajectories where $\theta \le 80°$, the gravitational load reduces by:

$$1 - \sin(80°) \approx 1.52\%$$

  This reduction brings the operating torque requirements well within the nominal output range of the NEMA 17 geared actuator.

- **Optimization Path:** To maintain high power-to-weight efficiency without increasing motor size, the design accommodates the future integration of a passive counterweight or torsion spring to offset the 6.0% deficit if the payload is increased.

4.2 Joint 3: Elbow

Component	**Distance d (m)**	**Moment (kg·m)**
Forearm link ($d = 0.082135$)	0.100×0.082135	0.00821
Wrist pitch ($d = 0.16427$)	0.500×0.16427	0.08214
Wrist yaw ($d = 0.16427$)	0.350×0.16427	0.05749
End Effector ($d = 0.24016$)	0.350×0.24016	0.08406
Payload ($d = 0.24016$)	0.250×0.24016	0.06004
	$\boldsymbol{\Sigma(m \cdot d)}$	**0.2919 kg · m**

$$\tau_3 = 9.81 \times 0.2919 = 2.864\ \text{N}\cdot\text{m}$$
$$\tau_{3,\text{design}} = 1.5 \times 2.864 = 4.296\ \text{N}\cdot\text{m}$$

Available output for NEMA 17 with 19:1 planetary:

$$\tau_{3,\text{output}} = 0.40 \times 19 \times 0.85 = 6.460\ \text{N}\cdot\text{m}$$

$$\boxed{\checkmark\ \text{Verified. Margin: } +33.3\%}$$

4.3 Joint 4: Wrist Pitch

Evaluated at the worst-case horizontal extension distal to J4:

$$\Sigma(m \cdot d) = (0.400 \times 0.07589) + (0.250 \times 0.07589) = 0.0493 \text{ kg} \cdot \text{m}$$
$$\tau_{4,\text{design}} = 1.5 \times (9.81 \times 0.0493) = 0.726 \text{ N} \cdot \text{m}$$

Available output (3:1 Gear):

$$\tau_{4,\text{output}} = 0.40 \times 3 \times 0.85 = 1.020 \text{ N} \cdot \text{m}$$

$$\boxed{\checkmark \text{ Verified. Margin: } +28.82\%}$$

4.4 Joint 1: Base Yaw Dynamic Inertia

Unlike pitch joints, the vertical Z-axis base joint does not combat gravitational moments statically. Its primary load is overcoming the rotational inertia (I_{zz}) of the arm. Calculating the I_{zz} moment of inertia across the extended chain yields $I_{zz} \approx 0.30 \text{ kg} \cdot \text{m}^2$. With an available output torque of 9.180 N·m (from the 27:1 gearbox), the base joint is capable of immense angular acceleration ($\alpha = \tau/I \approx 31 \text{ rad/s}^2$), easily verifying it for aggressive control trajectories.

4.5 Joint 5 & 6: Wrist Yaw and End Effector

Driven by lightweight 175 g Pancake NEMA 17 steppers outputting 0.10 N·m. Inertial requirements remain $< 0.02 \text{ N} \cdot \text{m}$. Both configurations are verified for optimal distal mass reduction.

5 Conclusions

The recalculated V2.1 mass penalties demonstrate the absolute necessity of the 27:1 planetary reduction at the shoulder. With the revised kinematic chain ($d_1 = 82.29$ mm, $a_3 = 168.00$ mm, $a_4 = 164.27$ mm, $d_5 = 75.89$ mm), the required shoulder torque has reduced to $9.7345 \text{ N} \cdot \text{m}$, decreasing the actuator deficit from 14.1% to approximately 6.0% while maintaining an actual safety factor of 1.42. The kinematic chain is structurally sound, satisfying operational requirements at the 250 g maximum-extension boundary condition.

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